Keshav Sharma

India | 8168289407 | keshavsharma71982@gmail.com | LinkedIn Link | GitHub Link | Portfolio

EDUCATION

Vellore Institute of Technology **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, C/C++, SQL, JavaScript, R
Machine Learning & Data Science: Deep Learning (LSTM Autoencoders), NLP (FinBERT, Transformers), Feature Engineering, Model Evaluation, Time-Series Analysis, Anomaly Detection, Probabilistic/Statistical Modeling (BG/NBD, Gamma-Gamma), Pandas, NumPy, Scikit-learn
Generative AI & LLMs: LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Google Gemini API, Vector Search/TF-IDF, Agentic AI Concepts, AI Coding Agents (GitHub Copilot, Cursor, Claude)
Backend & Frameworks: Flask, FastAPI, Node.js, React.js, REST APIs
Databases & Storage: MySQL, SQLite, Redis (NoSQL), SQL Query Optimization
Tools & Platforms: Git, GitHub, GitHub Actions, Docker, PyTest, CI/CD, VS Code, Jupyter Notebook, Power BI

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** ***Aug – Sep, 2025

- Applied Julius AI and LLM-based prompt engineering to analyze 2 real-world datasets (8,500+ records), extracting 15 structured business insights through natural language querying and AI-assisted data analysis.
- Iteratively refined prompts and validated AI-generated outputs for consistency, accuracy, and relevance, improving the reliability of generated insights while demonstrating effective use of AI-assisted workflows.
- Delivered a capstone project recognized with AICTE, VOIS, and Vodafone Idea Foundation certification, presenting findings to 3 mentors and translating AI-driven analysis into 4 actionable recommendations on user engagement and content trends.

PROJECTS

***Docwise* |LINK**

- Engineered Docwise, an AI-powered PDF Question Answering system using Google Gemini, enabling 95%+ query accuracy while reducing manual document search time by 40% through Retrieval-Augmented Generation (RAG) techniques.
- Implemented scalable document processing pipelines using TF-IDF vectorization, intelligent chunking, and real-time indexing for PDFs exceeding 200+ pages, improving responsiveness and retrieval performance.
- Applied prompt engineering, retrieval optimization, and hallucination-aware response validation to improve answer quality, while integrating chat history and session management in Streamlit to enhance usability and user retention by 35%.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built a full-stack expense management platform using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP-based parsing to automate receipt processing and improve processing time by 40%.
- Engineered backend APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export, enabling secure and reliable application workflows.
- Deployed the application across Render and Vercel, managing frontend-backend integration, authentication flows, database persistence, and production configuration for reliable end-to-end operation.

***Customer Lifetime Value Engine* |LINK**

- Developed an end-to-end probabilistic CLV pipeline using BG/NBD and Gamma-Gamma models on 1M+ transaction records, generating customer-level forecasts and segmenting customers into 5 value tiers for targeted business analysis.
- Designed Power BI dashboards to analyse cohort-level CLV, retention trends, and segment-wise revenue potential across 4,000+ customers and 12+ cohorts, enabling data-driven stakeholder reporting.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Built a real-time anomaly detection pipeline for NSE stock data using Isolation Forest and an LSTM Autoencoder, engineering 20+ time-series features and achieving a best F1-score of 0.87 across evaluated approaches.
- Integrated FinBERT sentiment analysis on financial news to contextualise price-volume anomalies, reducing false-positive alerts by 18% compared with market signals alone.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate |**LINK** 2026
- Prepare Data for ML APIs on Google Cloud Skill Badge |**LINK** 2026